Mini Project in Java
Assignment 01

Aim: Write a java program to showcase classes, object creation, methods, constructors using student information.

Objectives:

- To study classes objects and creation of objects of a class
- To study methods, return a value from the method, and pass parameters to the methods.
- To study constructors and use of constructors.
- To study different access specifies in java and their use for encapsulation.

Theory:

1) Define what is class and object with an example

A class is a blueprint or template that defines the properties and behaviours of an object.
An object is an instances of a class. It represents a real world entity and occupies memory.
Eg;
A student class can have attributes like name, roll number and marks. An object of the student class represents an individual student.

2) Methods in Java.

A method is a block of code that performs a specific task. Methods can accepts parameters and may returns a value syntax of method.

Eg: (i) access-specifier return-type method-name (parameters) {
//method body
return values;
}

(ii) Public int calculateGrade (int marks) {
return marks / 10;
}

(3) Write about the use of constructor with example.

A constructor is special method used to initialize objects

- it has the same name as the class
- it does not have a return type.
- automatically called when object in created.

```
Student (string name, int rollno) {
    this.name = name;
    this.rollno = rollno;
}
```

Conclusion:

Thus studied what are classes, object creation, methods constructors, access specifiers.

FAQs

1) What is a class and object?

A class is a blueprint or template use to define the properties and behaviours of objects. It does not occupy memory by itself. An object is an instance of class that represents a real world entity. When an object is created memory is allocated and the class member can be accessed using that object.

2) What is a constructor in java?

A constructor is a special method in java that is automatically called when an object of a class is created. It is mainly used to initialize the data member of the class. A constructor has the same name as the class and does not have a return type.

3) What is the default access specifiers and its scope?

The default access specifier when no access specifier is mentioned. In this case, the class members are accessible only within the save package. They cannot be accessed from outside the package.

4) What are different access specifiers?

Java provides four access specifiers :

(i) Private : accessible only within the same class.
(ii) Public : accessible from anywhere.
(iii) Protected : accessible within the same packages and subclasses.
(iv) Default : accessible within the same package.